

E1 — claim semantics

Method. Full reading of the EN manuscript; decomposition into 36 claim IDs (`CLAIM_MAP_AUDITOR.csv`) mapping each theorem/corollary/byproduct to literal hypotheses and Lean declarations; the author's `CLAIM_MAP.csv` was **not** consulted (it is in the internal-audit tree, deferred by §1.2).

Distinctions verified. All-orders additive b (Thm A, $b = N^2$, tower bound) vs eventual sharp $M(n)$ (Thm B) vs eventual maximum (1.2, witness optimal against unrestricted partitions); rooted defect s (Thm C: $c_4 \leq Q_s(n)$ eventually, maximum vs unrestricted partitions; no stability/classification claimed for $s > 0$); integral stability (Thm 6.1: hypothesis on order- ≤ 4 partitions, conclusion on graph edits 16δ); partition stability (Cor 6.1a: root fixed before an arbitrary, order-unrestricted partition; $\tau + 48\delta$ noncanonical pieces); extremal rigidity (Cor 6.2, both c_p and c_4); explicit tower threshold ($T(h+7)$, $b \leq T(h+8)$), explicitly impractical; triangular-only annex (8.1a) outside `import PaperIV`. The paper does not claim $b = 0$, a universal linear mixed gap, a practical threshold, or classification outside the chordal class (verified in §§6.3, 8.2, A.3 and the abstract).

Small orders / empty graphs / quantifier order. $n = 0$ and $n = 1$ are covered by Thm A (empty partition; Lean quantifies over all $n : \mathbb{N}$ with `Fin n`); $M(n) \geq n^2/6$ is used only for $n \geq 6$ (fails below — E5 negative control); Thm B/C, 6.1, 6.1a are eventual with the threshold chosen before G (and before δ , τ , Q); Cor 6.1a's root precedes every partition (checked in the Lean header). Theorem C's truncated subtraction in `defectTarget` is irrelevant for $n \geq s+1$.

Result. No semantic defect in the statements themselves. The attribution of Theorem C (E7-F1) is a bibliographic, not a semantic, defect. **E1 = PASS.**

Rendered from the Markdown semantic source with PyMuPDF; not compiled from the TeX file (no TeX engine installed).